

MCV 4U - Chapter 3: Derivatives and their Applications Quiz AM

Name [REDACTED]

Date _____

Total
16.5
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KNOWLEDGE

1. Determine the second derivative of the function, $f(x) = \frac{x^2+2x+1}{x+3}$. Fully simplify your answer.

$$f'(x) = \frac{(2x+2)(x+3) - (x^2+2x+1)}{(x+3)^2}$$

$$= \frac{2x^2 + 8x + 6 - x^2 - 2x - 1}{(x+3)^2}$$

$$= \frac{x^2 + 6x + 5}{(x+3)^2}$$

$$f''(x) = \frac{(2x+6)(x+3)^2 - (x^2+6x+5)2(x+3)}{(x+3)^4}$$

$$f''(x) = \frac{(x+3)[(2x+6)(x+3) - 2(x^2+6x+5)]}{(x+3)^4}$$

$$f''(x) = \frac{8}{(x+3)^3}$$

(3)

2. Find the equation of the tangent to the function $x^2y + y^4 = 4 + 2x$ at the point $(-1, 1)$.

$$(2xy + x^2 \frac{dy}{dx}) + 4y^3 \frac{dy}{dx} = 0 + 2$$

$$x^2 \frac{dy}{dx} + 4y^3 \frac{dy}{dx} = 2 - 2xy$$

$$\frac{dy}{dx} (x^2 + 4y^3) = 2 - 2xy$$

$$\frac{dy}{dx} = \frac{2 - 2xy}{x^2 + 4y^3}$$

$$\left. \frac{dy}{dx} \right|_{\substack{x=-1 \\ y=1}} = \frac{2 - 2(-1)(1)}{(-1)^2 + 4(1)^3} = \frac{4}{5}$$

$$y = \frac{4}{5}x + b$$

$$1 = -\frac{4}{5} + b$$

$$b = \frac{9}{5}$$

$$y = \frac{4}{5}x + \frac{9}{5}$$

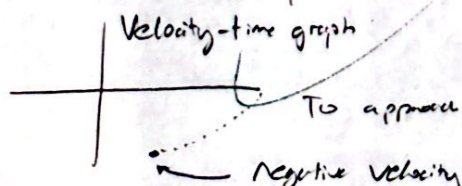
(4)

COMMUNICATION

1. What sign (+ or -) would the function $a(t)$ be if an object is slowing down with a negative velocity? Justify your response.

If the object is slowing down, that means velocity and acceleration are in opposite directions (opposite signs). That means acceleration is positive in this case.

It doesn't matter the position in this case, velocity must approach zero.



To approach 0, must be increasing, positive slope. $\therefore a(t)$ is positive.

(2)

APPLICATION

1. The position, at time t , in seconds, of an object moving along a horizontal line is given by $s(t) = \frac{4}{3}t^3 + 6t^2 - 40t$ for $0 \leq t \leq 8$.

a) When is the object at rest?

$$\begin{aligned} s'(t) = v(t) &= 4t^2 + 12t - 40 \\ &= 4(t^2 + 3t - 10) \\ &= 4(t+5)(t-2) \end{aligned}$$

$\therefore$ Object is at rest at 2 seconds.

(2)

b) What is the velocity at 3 seconds?

$$\begin{aligned} v(3) &= 4(3)^2 + 12(3) - 40 \\ &= 32 \text{ m/s} \end{aligned}$$

$\therefore$ At 3 seconds, object is moving at a velocity of 32 m/s

(1)

c) What is the acceleration when the velocity is $32 \frac{\text{m}}{\text{s}}$?

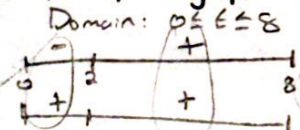
$$\begin{aligned} s''(t) = v'(t) = a(t) &= 8t + 12 \\ a(3) &= 8(3) + 12 \\ &= 36 \text{ m/s}^2 \end{aligned}$$

Velocity is 32 m/s at 3 seconds
 $\therefore$ acceleration is 36 m/s^2

(2)

d) intervals where object is speeding up/slowing down

$$\begin{aligned} v(t) &= 4(t+5)(t-2) \\ a(t) &= 4(2t+3) \end{aligned}$$



Speeding up from $(2, 8)$
Slowing down from $(0, 2)$

1.5 (2)

2. A swimming pool is treated to control the growth of bacteria. Suppose that t days after a treatment, the number of bacteria in thousands per cubic centimeter is $N(t) = t^{\frac{3}{2}} - 3t^{\frac{5}{2}} + 3$. Determine the lowest and highest number of bacteria over the time interval of $0 \leq t \leq 1$. Round to 3 decimal places.

$$\begin{aligned} N(t) &= t^{\frac{3}{2}} - 3t^{\frac{5}{2}} + 3 \\ 0 &\leq t \leq 1 \end{aligned}$$

$$\begin{aligned} N'(t) &= \frac{3}{2}t^{\frac{1}{2}} - \frac{15}{2}t^{\frac{3}{2}} \\ &= \frac{3}{2}t^{\frac{1}{2}}(1 - 5t) \end{aligned}$$

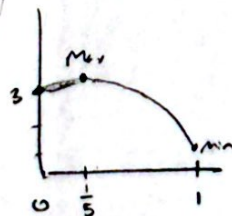
$$\therefore t = 0, \frac{1}{5}$$

$$N(0) = 3$$

$$N\left(\frac{1}{5}\right) = 3.036$$

$$N(1) = 1$$

$$\begin{aligned} &> \left(\frac{1}{5}\right)^{\frac{3}{2}} - 3\left(\frac{1}{5}\right)^{\frac{5}{2}} + 3 \\ &= 0.089 - 0.054 + 3 \\ &= 3.036 \end{aligned}$$



Therefore the highest amount of bacteria is 3036 per cm^3 at 0.2 days.
The lowest amount of bacteria is 1000 per cm^3 at 1 day.

(3)